Record - Week 03

Topic: Node.js and Express.js

Name: _____ Roll No: _____ Date: _____

Aim

To build a REST API server using Node.js and Express with routing and middleware.

Software Requirements

Node.js 18+, npm, VS Code, Postman

Procedure

1. Run npm init and install express.
2. Create server.js with JSON parsing and a logger middleware.
3. Create routes/api.js with GET and POST routes for students.
4. Mount the router at /api.
5. Start the server and test the endpoints in Postman.

Program

server.js

```
const express = require("express");
const apiRoutes = require("./routes/api");

const app = express();
const PORT = process.env.PORT || 3000;

app.use(express.json());

// simple logging middleware
app.use((req, res, next) => {
  console.log(`${req.method} ${req.url}`);
  next();
});

app.get("/", (req, res) => res.send("Express server is running"));
app.use("/api", apiRoutes);

app.listen(PORT, () => console.log(`Server listening on port ${PORT}`));
```

routes/api.js

```
const express = require("express");
const router = express.Router();

let students = [
  { id: 1, name: "Asha", branch: "CSE" },
  { id: 2, name: "Ravi", branch: "IT" },
];

router.get("/students", (req, res) => res.json(students));

router.get("/students/:id", (req, res) => {
  const s = students.find((x) => x.id === Number(req.params.id));
  if (!s) return res.status(404).json({ error: "Student not found" });
  res.json(s);
});

router.post("/students", (req, res) => {
  const { name, branch } = req.body;
  if (!name || !branch) return res.status(400).json({ error: "name and branch required" });
  const student = { id: students.length + 1, name, branch };
  students.push(student);
  res.status(201).json(student);
});

module.exports = router;
```

package.json

```
{
  "name": "week-03-node-and-express",
  "version": "1.0.0",
  "description": "Express server with REST routes",
  "main": "server.js",
  "scripts": {
    "start": "node server.js",
    "dev": "node --watch server.js"
  },
  "dependencies": {
    "express": "^4.19.2"
  }
}
```

Output

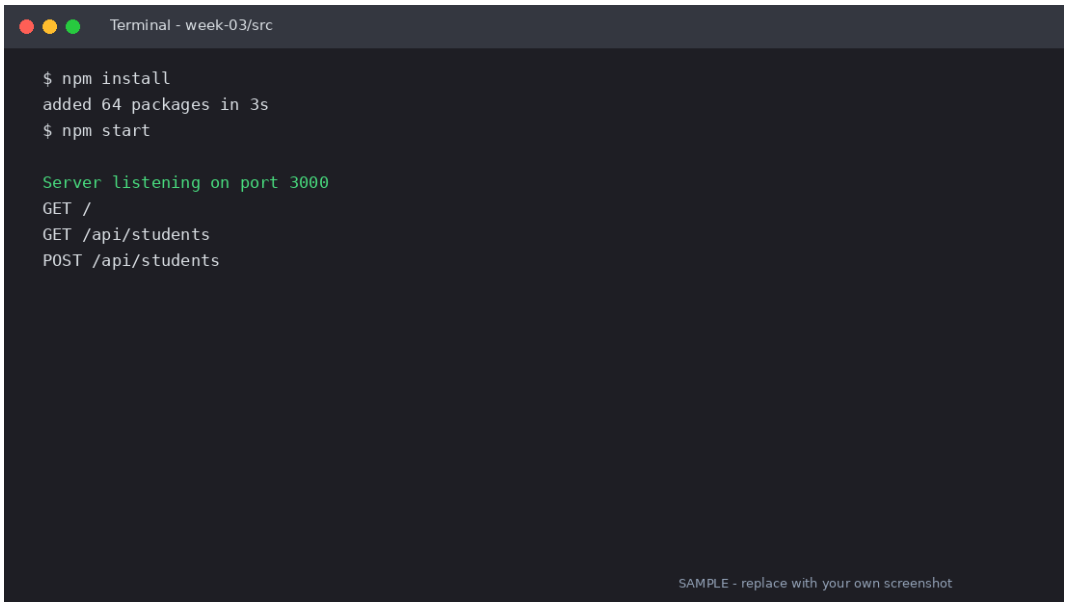

Fig: express server terminal

Fig: postman api response

Result

The program was executed successfully and the expected output was obtained.